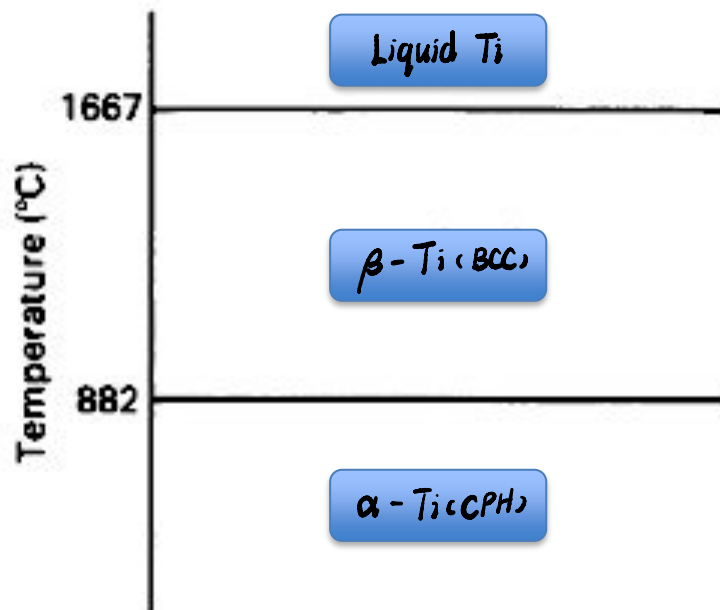


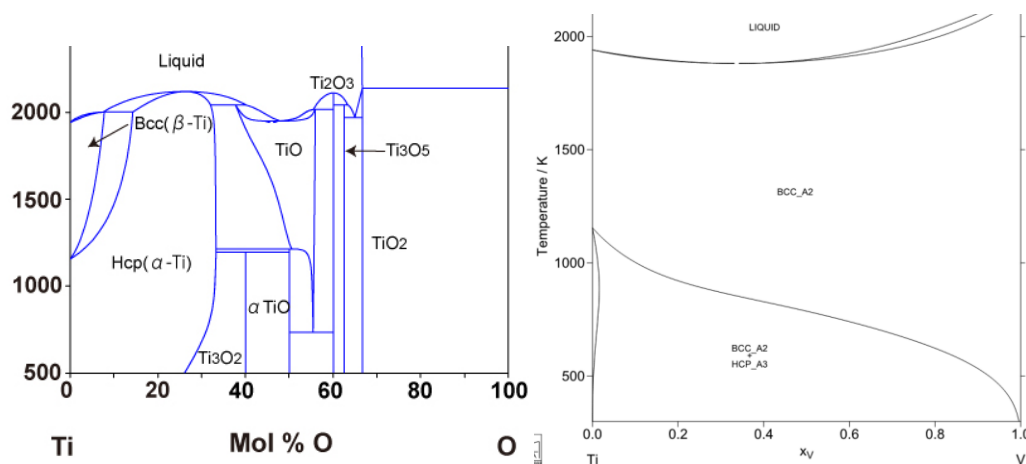
2. Alloys: Ti

1. Titanium (Ti)

Ti is a lightweight metal with many advantages for engineering application. Ti is a polymorphic metal. Label the different phases of Ti at different temperatures and indicate their lattice packing in the figure below.



By adding different alloy elements, the phase transformation temperature can be changed. According to the phase-diagram (below left) of Ti-O, O is an α stabiliser; while the phase-diagram of Ti-V (below right) indicates that V is a β stabiliser. Apart from O, C and N can also act as α stabiliser. These small atoms form interstitial solid solutions, interstitial solid solution strengthening is much more effective than substitutional solid solution strengthening.



2. Ti alloys

Fill the table below accordingly.

Ti alloys	α alloy	Near α alloy	α - β alloy	Near β alloy	β alloy
Example alloys	C.P. Ti-5Al-2.5Sn	Ti-6Al-2Sn-4Zr-2Mo	<u>Ti 64</u>	Ti-10V-2Fe-3Al	Ti-13V-11Cr-3Al
Equilibrium constitution at T_{room}	<u>Single-phase alloy α-Ti</u>	Engineering interesting area is $\alpha + \beta$	<u>$\alpha + \beta$</u>	$\alpha + \beta$	<u>β with few% α</u>
Properties	Not engineering relevant	Best <u>creep resistance</u> among all Ti alloys	A good balance of <u>fatigue</u> and <u>creep resistance</u> at elevated T	Very good strength <u>at low temperature</u>	Highest strength Ti alloy
Strengthening mechanisms *	<u>SS, GB, WH</u>	SS; GB	SS; GB; Ppts	<u>SS, GB, ppts</u>	SS; GB; ppts
Application	Not engineering relevant	<u>Aeroengine compressor blade</u>	<u>Fan blade of aeroengine</u>	<u>Landing gear</u>	/////

- SS: solid solution strengthening;
- GB: grain boundary strengthening;
- WH: work hardening;
- Ppts: Precipitation strengthening.

Learn more detailed explanation in the lecture slides.

3. Summary of aerospace Ti alloys

